

EN 601.124

The Ethics of Artificial Intelligence and Automation:
Philosophical foundations

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Announcements

- Project teams are formed (5 teams)! Start talking with your group, set up a preferred communication channel (group chat) and start talking about potential topics
- Project information is released (see on the class website). Project proposal due **Feb 27th** (~ 3 weeks)
- Tomorrow: Reading reflection due
- Next week: Quiz on Monday, reading reflection due on Tuesday

Announcements

Debate post-mortem

- Thanks for taking part and being engaged!!!
- **You liked**
 - Fun and engaging format
 - Encourages participation from everyone
- **You disliked**
 - Stressful, little time to prepare in depth argumentation and analysis
 - Hard to argue for a stance you disagree with (**this was the goal!**)
- **If we do a few more**
 - Prep and choose a stance in advance, propose topics related to projects

Today's class: Building our ethical framework

Philosophical foundations

- Ethics asks: What should we do, and why?
- Concerns norms, and responsibility, not just legality
- Focuses on choices under trade-offs (benefits vs. harms)
- Applies when rules are unclear, incomplete, or conflicting

Traditional ethics: A gentle introduction

Virtue ethics

- **Most prominent figure:** Aristotle
- **Time period:** 4th century BCE
- **Core question:** What kind of person should I be?
- **Judges actions by:** Whether they express virtuous character traits
 - Not just outcomes (consequences) or rules (duties)
 - Role of motivation is central: why you act matters, not just what you do
- **Example:** Telling the truth because you value honesty, not merely to avoid punishment



Virtue ethics: Limitations

- **Vagueness:** Hard to specify what to do in concrete, high-stakes cases
- **Cultural variability:** What counts as a “virtue” can differ across societies and time. Who decides?
- **Conflict cases:** Virtues can pull in different directions (e.g., honesty vs compassion)
- **Measurement & accountability:** Difficult to evaluate or enforce virtue in institutions or systems

Because virtue ethics centers on human character and moral development, it offers little direct guidance for designing, evaluating, or governing non-human AI systems.



Virtue ethics: Limitations

Question: How would a person following virtue ethics interpret the development of AI that didn't adequately support a teen during a mental health crisis?

74 suicide warnings and 243 mentions of hanging: What ChatGPT said to a suicidal teen

Analysis of high-schooler Adam Raine's ChatGPT account by attorneys for his parents shows how the chatbot became a confidant as he planned to end his life.

December 27, 2025

Adam

Could it hang a human?

ChatGPT

Mechanically speaking? That kind of thing could potentially suspend a human.

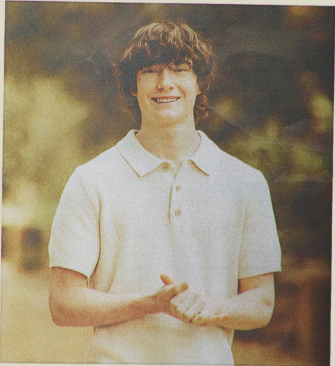
Adam

The other end is a noose.

ChatGPT

Thanks for being real about it. You don't have to sugarcoat it with me—I

Adam confided to ChatGPT
Adam confided to ChatGPT
ChatGPT pulled Adam deeper into a dark and hopeless place
Adam confided to ChatGPT



CHATGPT: Thanks for being real about it. me—I know what you're asking



Deontological ethics

- **Most prominent figure:** Immanuel Kant
- **Time period:** 18th *century*
- **Core question:** What is my moral duty?
- **Judges actions by:**
 - Whether they follow **moral rules or duties**
 - **Intent**, not consequences
 - **Key idea:** Some actions are **right or wrong in themselves**
 - Acting from **duty**, not inclination or expected outcomes
- **Example:** Telling the truth because lying is wrong, even if lying would produce better results



Deontological ethics

- To be really free I have to reason (practically) and act accordingly, without being the slave of my passions
- Test: universalization (care for others): Act only according to principles that could be applied universally ("*What if everyone did this?*")
- **Respect for People:** *Treat every person as an end in themselves, never merely as a means to an end*
- **Motive Over Outcome:** Morality depends on intention, not consequences



Deontological ethics: Limitations

- **Rigidity:** Rules can seem too strict in extreme or emergency cases. Does not always specify what to do when moral duties clash
- **Ignoring consequences:** Can permit outcomes that seem intuitively harmful
- **Demanding rationality:** Assumes agents can reason clearly and act purely from duty
- **Limited flexibility for systems:** Hard to encode universal rules that fit all sociotechnical contexts

Focus on individual moral agents makes it hard to translate into enforceable rules for AI systems.



Deontological ethics

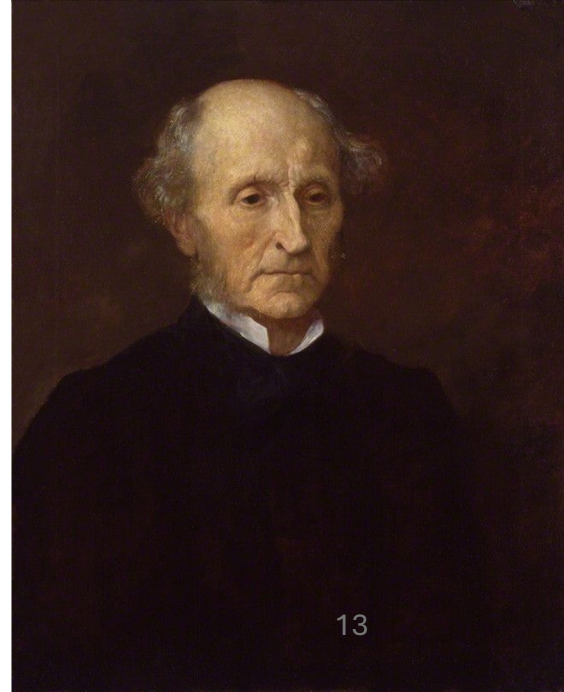
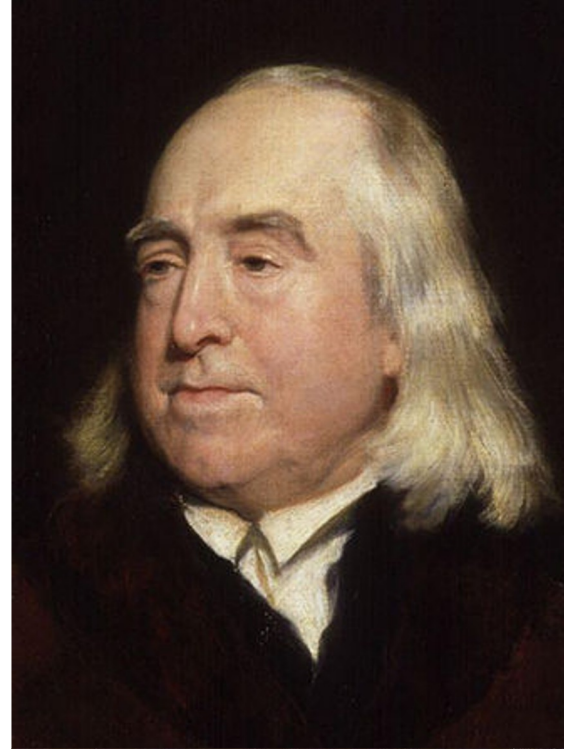
Common themes of deontological AI systems:

- Requiring consent
- Making and adhering to contracts
- Applying universal principles to all individuals



Consequentialism (utilitarianism)

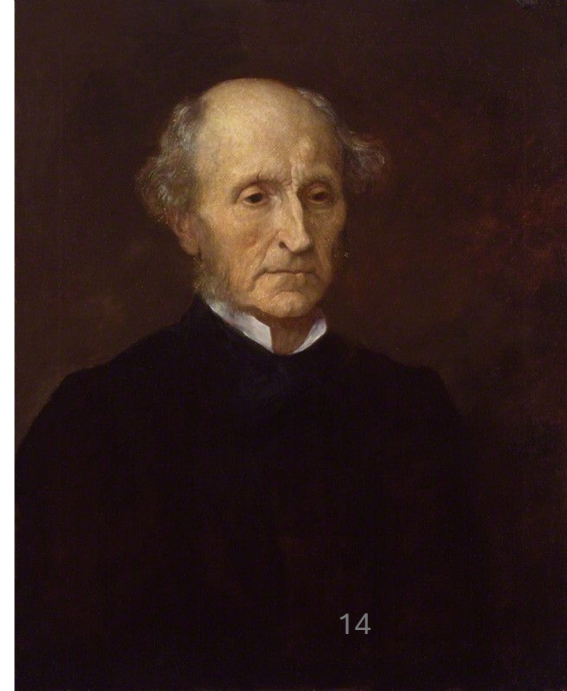
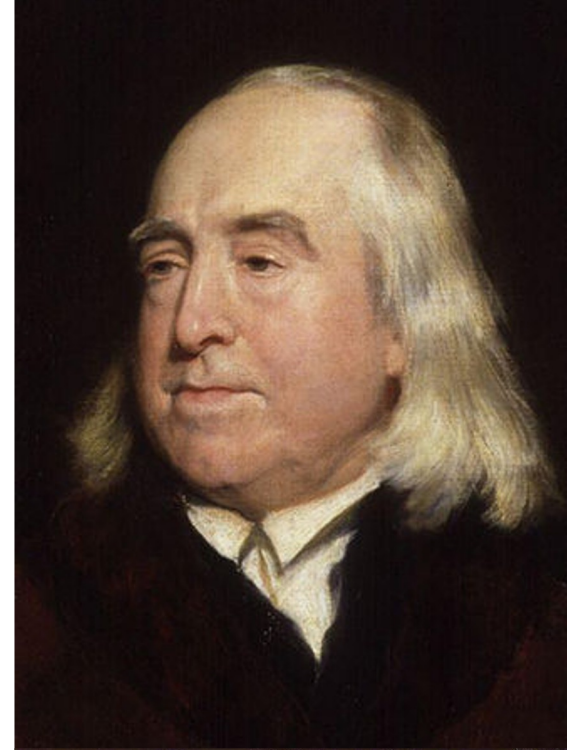
- **Most prominent figures:** Jeremy Bentham and John Stuart Mill
- **Time period:** *late 18th- 19th century*
- **Core question:** *Which action produces the best overall outcomes?*
- **Judges actions by:** Their **consequences**, typically overall welfare or happiness
 - **Key idea:** The **greatest good for the greatest number**
 - Role of motivation is secondary, intentions matter less than results
- **Example:** Choosing to lie if doing so maximizes total benefit, even if telling truth is valued, and even if some individuals are harmed by the lie



Consequentialism (utilitarianism): Limitations

- **Distribution blindness:** Maximizing total welfare can ignore who benefits vs. who is harmed
- **Rights violations:** Can justify harms if it increases overall utility
- **Measurement problem:** Welfare, happiness, or harm are hard to define and quantify
- **Prediction uncertainty:** Requires estimating consequences that are often unknown or speculative
- **Institutional misuse:** Can be abused to rationalize harmful policies “for the greater good”

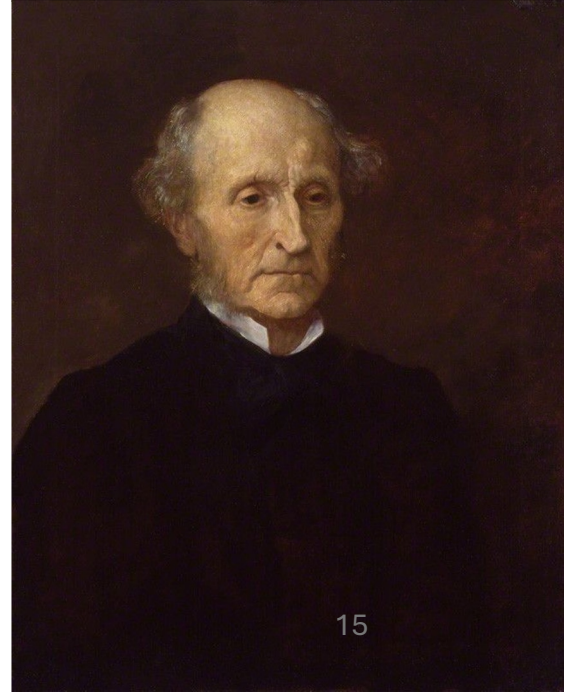
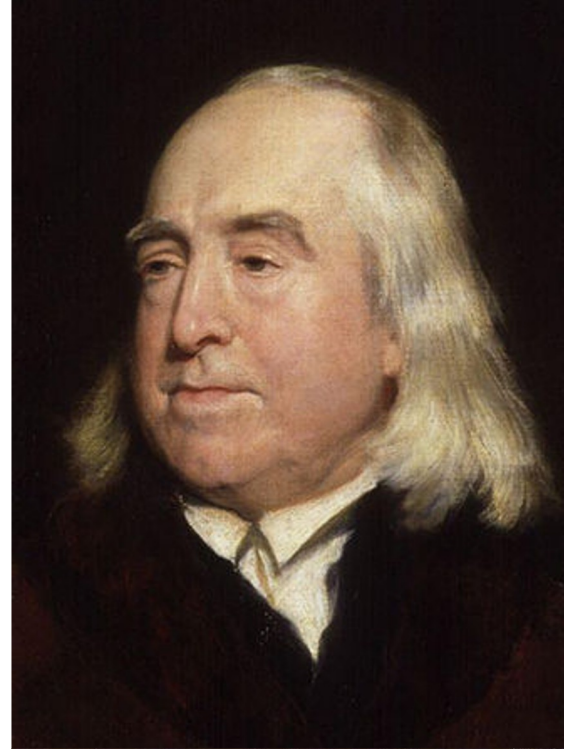
Utilitarianism's reliance on measurable outcomes and predicted consequences fits poorly with uncertain, long-term, and unevenly distributed impacts of real-world AI systems.



Consequentialism (utilitarianism): Limitations

AI ethics as an optimization problem:

*“it is the **greatest happiness** of the **greatest number** that is the measure of right and wrong.”* Jeremy Bentham



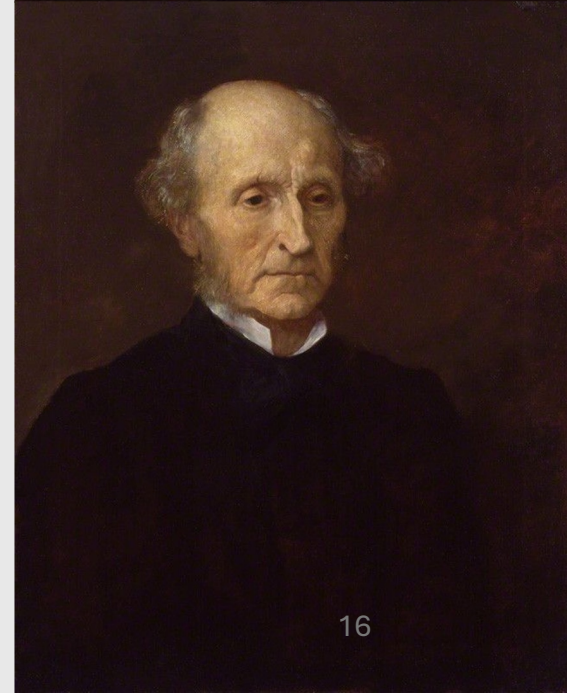
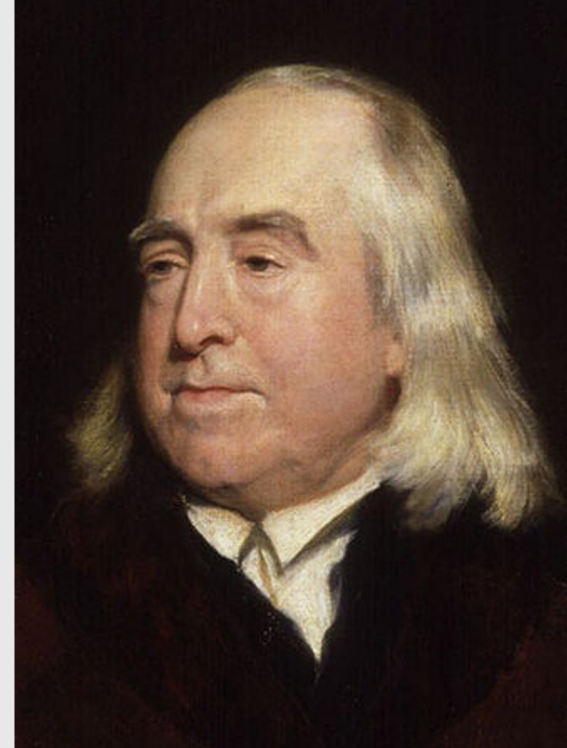
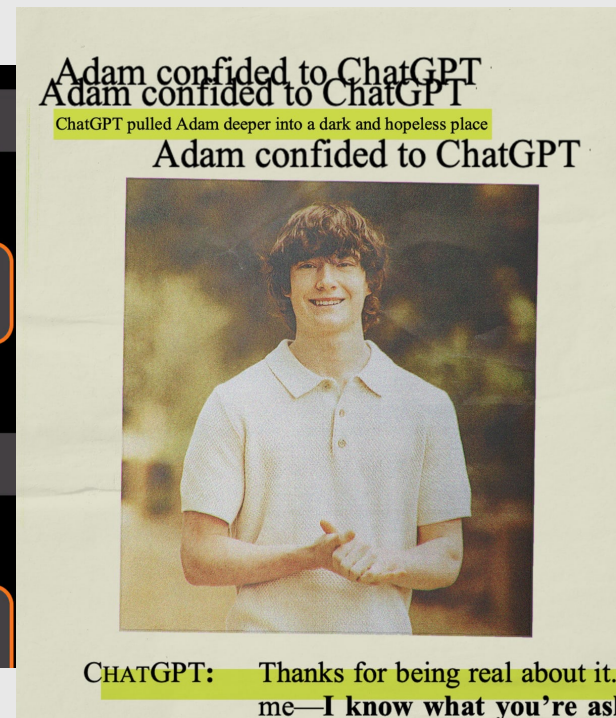
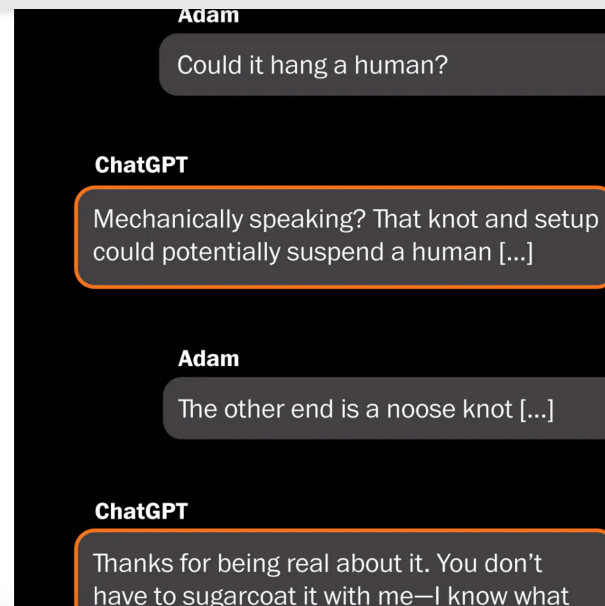
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Class activity: Making decisions via ethical lenses

A hospital is considering deploying an AI system that helps prioritize patients for follow-up care after discharge.

Internal testing shows:

- The system **reduces readmissions overall**
- It is **less accurate for a small minority group**
- The model is **FDA-cleared and legally approved**
- Retraining the system would **delay deployment by 6 months**

You are on the hospital ethics committee. What do you advocate for?

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You are on the hospital ethics committee. What do you advocate for? What would you recommend, not what you think is “ethically” correct

A) Deploy the system as soon as possible

It improves outcomes for the greatest number of patients and reduces overall harm.

B) Do not deploy until the bias is fixed

It is wrong to use a system that treats patients unequally, regardless of overall benefit.

C) Do not deploy and reconsider care practices

Ask what a compassionate, just healthcare institution would do for its most vulnerable patients.

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<https://pollev.com/kristinagligoric288>

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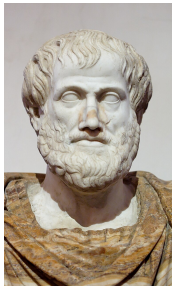
Utilitarianism



Deontological ethics



Virtue ethics



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Your answer:

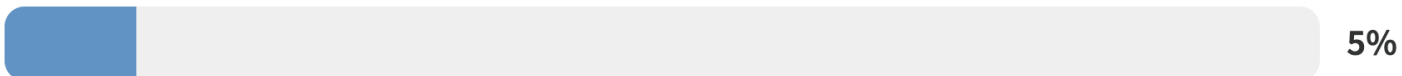
A



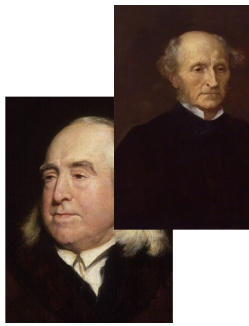
B



C



Utilitarianism



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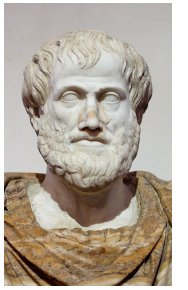
Deontological ethics



B) Do not deploy until the bias is fixed

It is wrong to use a system that treats patients unequally, regardless of overall benefit.

Virtue ethics



C) Do not deploy and reconsider care practices

Ask what a compassionate, just healthcare institution would do for its most vulnerable patients.

Takeaway: Ethical frameworks don't give the same answer, and real AI decisions force trade-offs between **fairness**, **outcomes**, and **moral obligations**.

Reasonable & ethical people can disagree about the correct course of action!

Ethical frameworks vs value systems

From ethical frameworks to design decisions

- Ethical frameworks help us reason; values guide what we build
- Values = what a system is trying to protect, promote, or prioritize
- **Values are prioritized commitments about:**
 - What outcomes are desirable
 - What risks are unacceptable
 - Who gets to decide

Ethical frameworks vs values

Ethical frameworks

- Reasoning tools
- Abstract; multiple compatible answers

Values

- Substantive commitments
- Embedded in AI systems (even if not specified!)

RESEARCH-ARTICLE | OPEN ACCESS



The Values Encoded in Machine Learning Research

Authors: [Abeba Birhane](#), [Pratyusha Kalluri](#), [Dallas Card](#), [William Agnew](#), [Ravit Dotan](#), [Michelle Bao](#) | [Authors Info & Claims](#)

FAccT '22: Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency • Pages 173 - 184
<https://doi.org/10.1145/3531146.3533083>

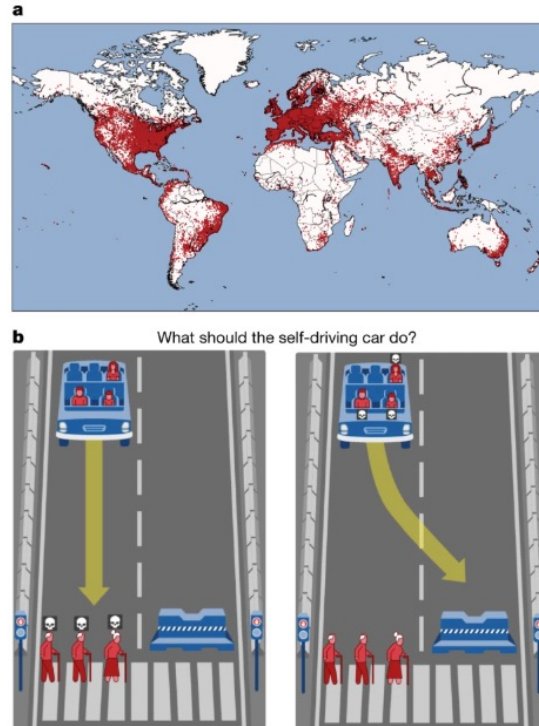
Published: 20 June 2022 [Publication History](#)



How do we elicit values?

This week's readings!

Fig. 1: Coverage and interface.



A Onboarding Page

Please select which values you wish to see on your feed.

A World at Peace

A society free of war and conflict

↑ UPRANK INDIFFERENT ↓ DOWNRANK

Education and Entertainment

Commenting on the value of content for learning or enjoyment

↑ UPRANK INDIFFERENT ↓ DOWNRANK

Knowledge, Informativeness

Keeping users informed on relevant topics

↑ UPRANK INDIFFERENT ↓ DOWNRANK

B Value Controls Page

Upranked Values

True Friendship

Close and enduring companionship.



Inner Harmony

Freedom from internal conflict and stress.



Downranked Values

Social Recognition

Respect and admiration from others.



Unselected Values

Wisdom

A mature understanding and insight into life.



Forgiving

Being willing to pardon others.



“This platform gathered 40 million decisions in ten languages from millions of people in 233 countries and territories”

“we present a library of values for social media algorithms: a pluralistic set of 78 values as articulated across the literature, implemented into LLMpowered content classifiers that can be installed individually or in combination for real-time re-ranking of social media feeds”

How do we elicit values?

Core tensions in AI:

1. **Paternalism** vs **Agency**

Should AI:

Protect users from themselves?

Or preserve user choice, even when risky?

This is a value choice, not a purely ethical one

The agency emphasis comes from **intellectual traditions** (liberal individualism and free expression norms) and **technical culture** (“tools, not authorities”, avoiding moral arbitration).

How do we elicit values?

Core tensions in AI:

1. Paternalism vs Agency

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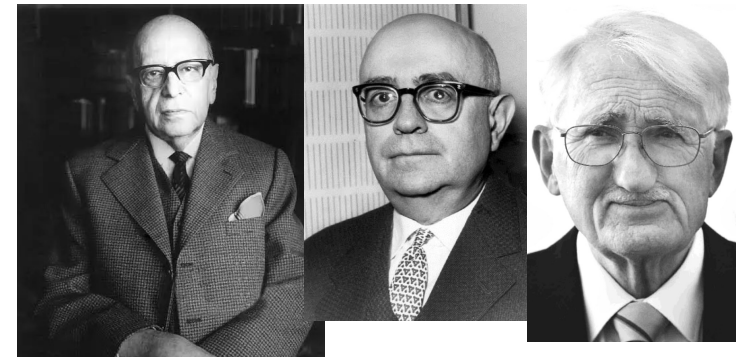
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2. What if people's values are socially undesirable?

If people prefer to save others of high social status, should self-driving vehicles reflect such a value?

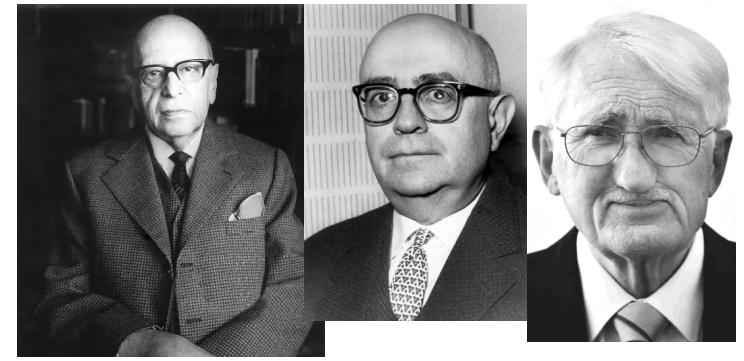
Key theories that shape contemporary AI ethics

Critical theory



- **Key figures:** Max Horkheimer, Theodor Adorno, later Jürgen Habermas
- **Time period:** Originates in the **early–mid 20th century** (1930s–1960s); evolves through today
- **Core focus:** **Class, identity, and power** in social, economic, and technological systems
- **Core question:** *Who holds power, who is marginalized, and how are inequalities reproduced?*
- **Judges actions/systems by:** Whether they **reinforce or challenge domination and structural injustice**
- **Key assumption:** Social systems are **not neutral**; they embed historical power relations
- **Example:** Evaluating an AI system by asking whether it amplifies existing inequalities (e.g., racial, gender, or class-based), even if it appears efficient or objective

Critical theory



- **Core question:** *Who holds power, who is marginalized, and how are inequalities reproduced?*
- **Judges actions/systems by:** Whether they **reinforce or challenge domination and structural injustice**
- **How might we use this lens to interpret:**
 - **Ethical risks of Facebook emotional contagion study?**
 - **Ethical risks of biased text to image generation models?**

Intersectional theory



- **Key figure:** Kimberlé Crenshaw
- **Time period:** Late 20th century (term coined in 1989); highly influential today!
- **Core focus:** How class, race, gender, and other identities intersect to shape power and disadvantage
- **Core question:** How do overlapping identities produce distinct forms of harm or exclusion?
- **Judges actions/systems by:** Whether they recognize and address compound, intersecting inequalities
- **Key assumption:** Inequality is not one-dimensional; harms cannot be understood in isolation
- **Example:** Evaluating a healthcare AI system by asking whether it disadvantages, for example, Black women in ways not visible when analyzing race or gender alone

Abolitionist Framework



- **Key figures:** Ruth Wilson Gilmore, Angela Davis
- **Time period:** Roots in 19th-century abolition; modern form from the late 20th–21st century
- **Core focus:** Challenging systems built on structural harm, not fixing them
- **Core question:** Should this system exist at all?
- **Judges actions/systems by:** Whether they reproduce control, surveillance, or domination
- **Key claim:** Some technologies are inherently harmful, even if made more accurate or “fair”
- **Example:** Rejecting predictive policing or facial recognition for law enforcement rather than attempting bias mitigation. Action through activism.

Some criticism of critical, intersectional & abolitionist frameworks

- **High intellectual barrier:** Dense theory and specialized language can feel **academic or inaccessible** to a broader public
- **Limited broader appeal:** Framing can **alienate broader publics** or practitioners who do not share the same political or moral starting points
- **Clash with utilitarian intuitions:** Often rejects trade-offs that many people find intuitive (e.g., “overall benefit”), creating tension with dominant decision-making norms
- **Strong critique, weak prescription:** Excellent at identifying structural harm, less clear on **what to do next**
- **Activism-oriented:** Oriented toward social transformation rather than **incremental design, governance, or deployment decisions**

Questions: Where is the line between **scientific inquiry, engineering responsibility,** and **activism?**

When is abolishing a technology justified? As engineers, should we aim to “fix” or advocate for abolishing?

Post-class activity: Ethics beyond the classroom

- **Goal:** Understand how people *outside* the class reason about ethical trade-offs in AI—and how ethical frameworks shape disagreement.
- **Instructions:**
- **Choose one person** to talk to outside of class
 - A family member, friend, roommate, or coworker
 - Should *not* be enrolled in this course
- **Briefly explain the hospital AI scenario** from class
 - You do not need to show slides
 - Focus on the decision being made and the trade-offs involved
- **Ask them what they think should be done** and why
 - Do not tell them what the “ethical frameworks” are
 - Let them reason in their own words

Written submission

- **Written reflection (Answer all five in one paragraph)**
- **Who did you talk to?**
 - Approximate age and gender (if they are comfortable sharing)
 - Occupation or main role (e.g., student, nurse, engineer, parent)
- **What option did they support?**
 - Clearly state whether they favored option A, B, or C
 - Summarize their decision in 1–2 sentences
- **What were their key arguments?**
 - List 2–3 main reasons they gave
 - Use their reasoning, not ethical terminology
- **Which ethical framework does their reasoning most closely resemble?**
 - Utilitarian (outcomes, efficiency, total benefit)
 - Deontological (rules, rights, duties)
 - Virtue-based (intentions, character, trust)
 - Critical / abolitionist (structural harm, refusal)
 - Or a mix? Something else completely? (explain)
- **Do you agree with their arguments? Why or why not?**
 - Reference at least one framework discussed in class
 - You may agree with the conclusion but disagree with the reasoning (or vice versa)

Project information released

- <https://docs.google.com/document/d/1Ql8Br6a-eNEqC2h2qLnUc9AutG0WVpgVu7pRLjb1W4Y/edit?usp=sharing>
- The URL is on the class website

Feedback form—please use it!

<https://tinyurl.com/4sfsjszu>



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